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Department of Computer Science and Engineering

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Link Layer and LAN

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Unit – 4 Link Layer and LAN Roadmap



- Introduction
- Error detection, correction
- Multiple access protocols
- Switched LANs
 - LL addressing, ARP
 - Ethernet
 - switches
- A day in the life of a web request
- Physical layer
- Wireless LANs: IEEE 802.11

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MAC Addresses

- 32-bit IP address:
 - *network-layer* address for interface
 - used for layer 3 (network layer) forwarding
 - e.g.: 128.119.40.136
- MAC (or LAN or physical or Ethernet) address:
 - function: used “locally” to get frame from one interface to another physically-connected interface (same subnet, in IP-addressing sense)
 - 48-bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
 - e.g.: 1A-2F-BB-76-09-AD

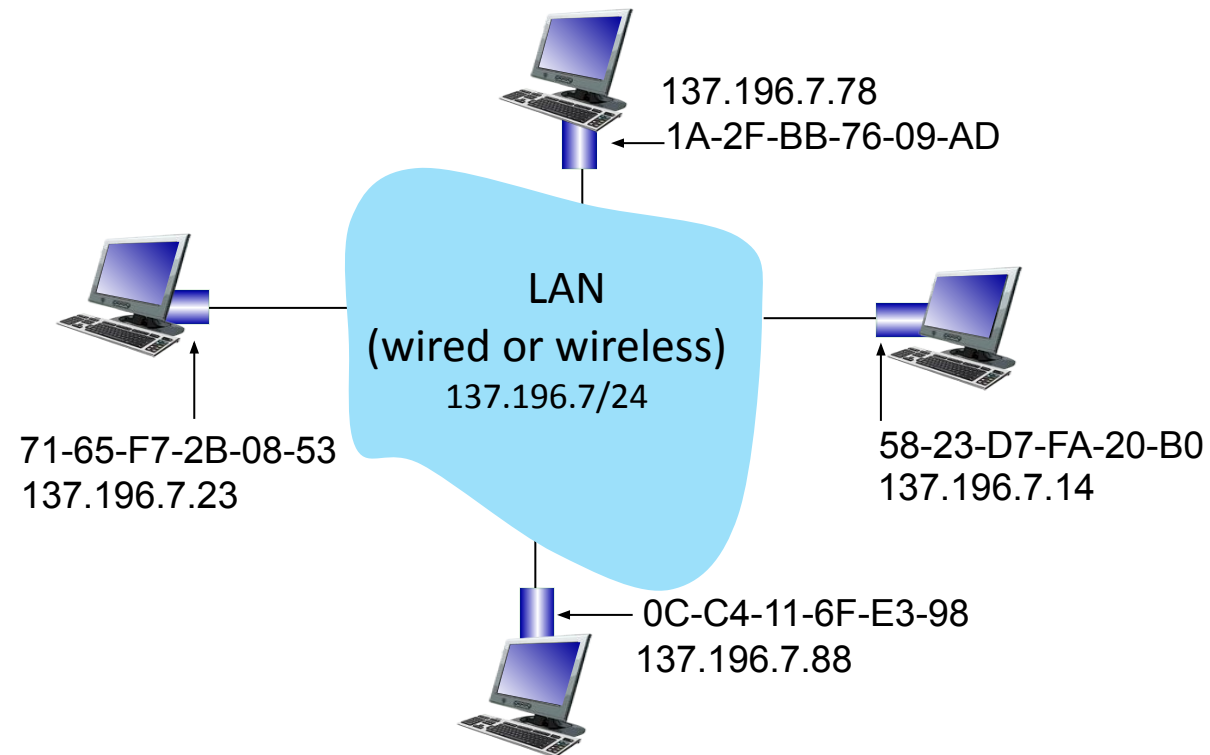
— hexadecimal (base 16) notation
(each “numeral” represents 4 bits)

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MAC Addresses

Each interface on LAN

- has unique 48-bit **MAC** address
- has a locally unique 32-bit IP address (as we've seen)



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MAC Addresses

- MAC address allocation administered by IEEE
- manufacturer buys portion of MAC address space (to assure uniqueness)
- analogy:
 - MAC address: like Social Security Number
 - IP address: like postal address
- MAC flat address: portability
 - can move interface from one LAN to another
 - recall IP address *not* portable: depends on IP subnet to which node is attached

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ARP: Address Resolution Protocol

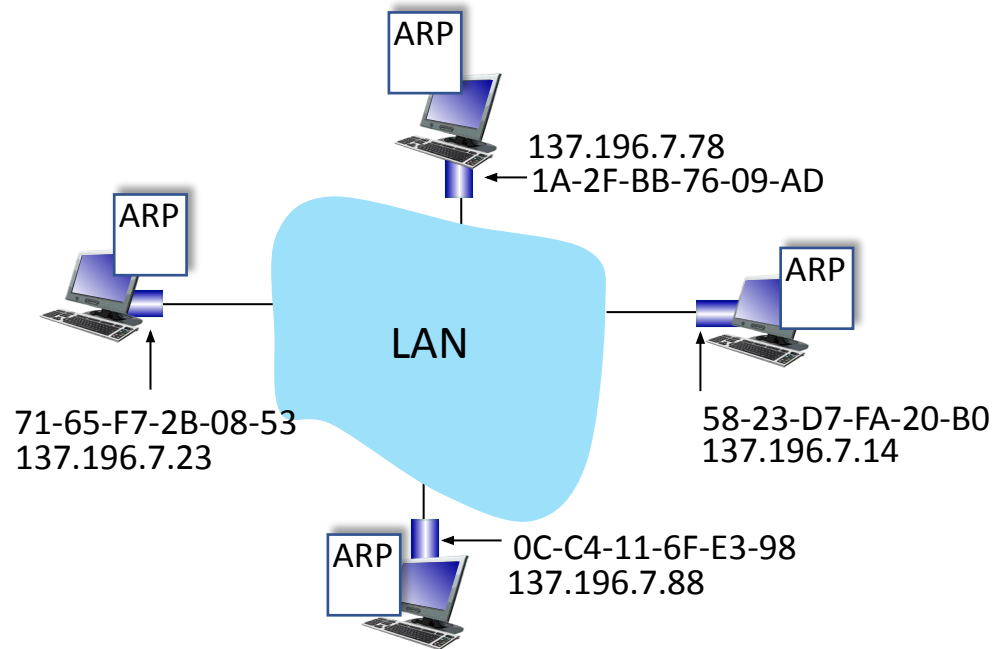
Question: how to determine interface's MAC address, knowing its IP address?

ARP table: each IP node (host, router) on LAN has table

- IP/MAC address mappings for some LAN nodes:

< IP address; MAC address; TTL >

- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)



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ARP Protocol in action

Example: A wants to send datagram to B

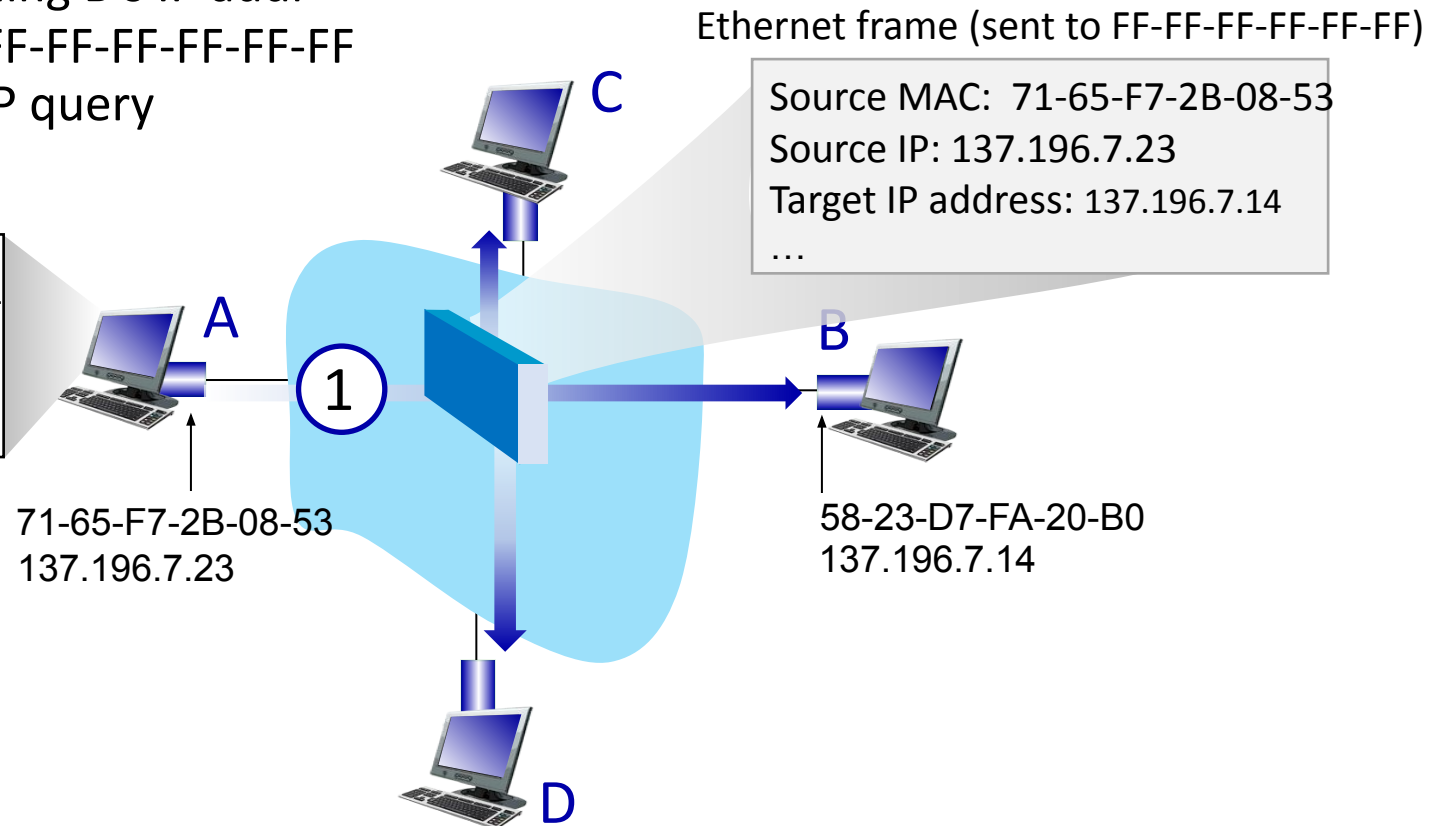
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address

A broadcasts ARP query, containing B's IP addr

- ①
- destination MAC address = FF-FF-FF-FF-FF-FF
 - all nodes on LAN receive ARP query

ARP table in A

IP addr	MAC addr	TTL

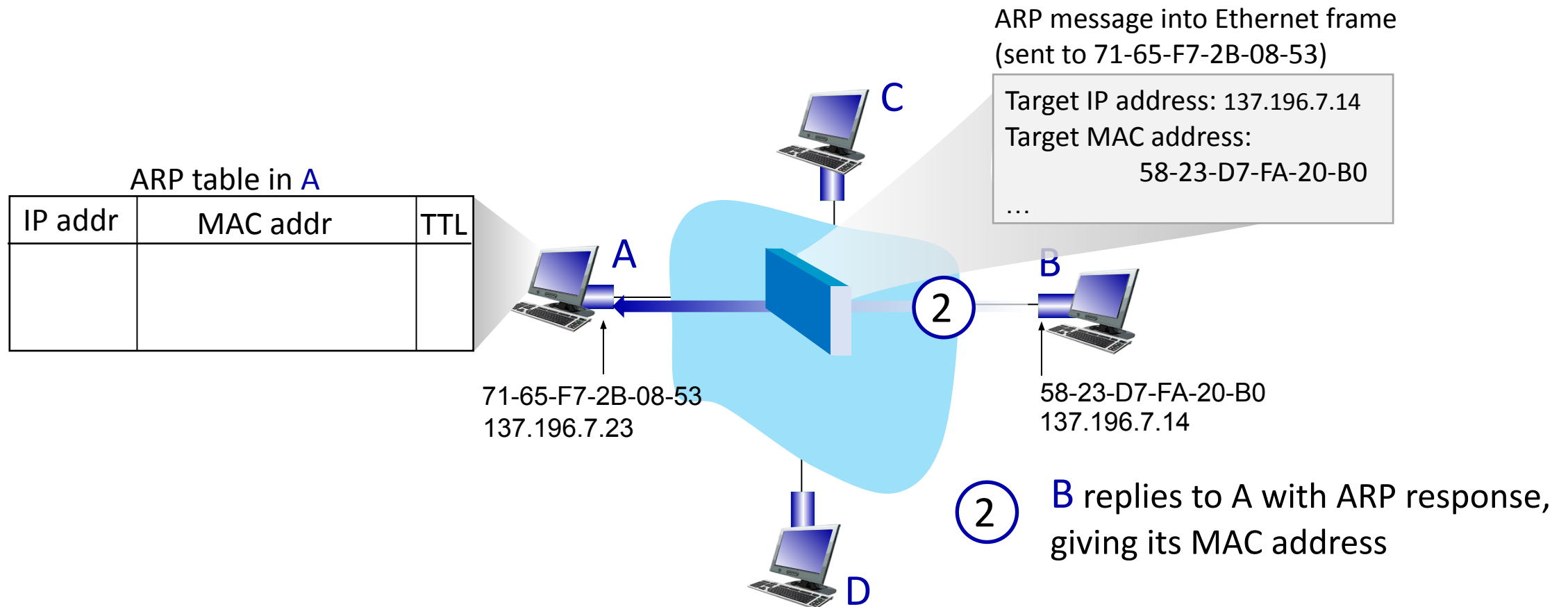


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ARP Protocol in action

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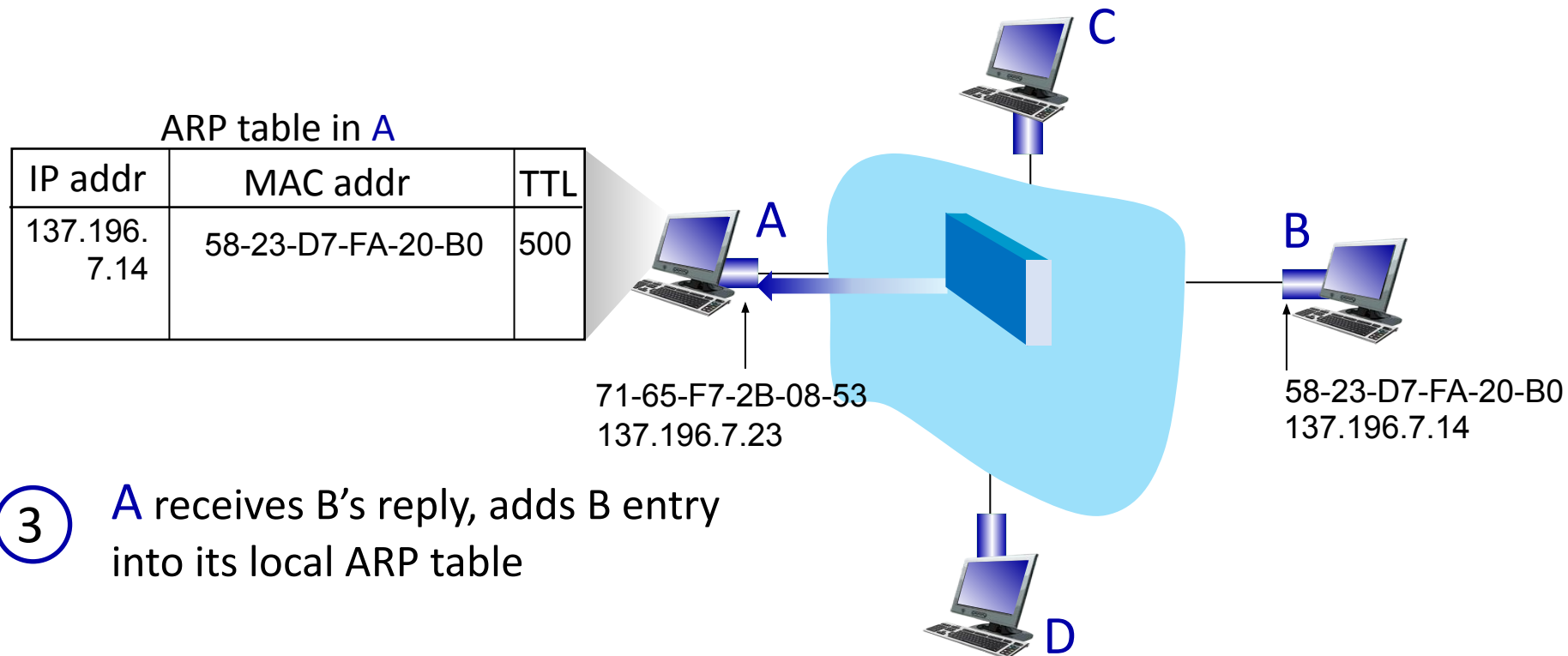


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ARP Protocol in action

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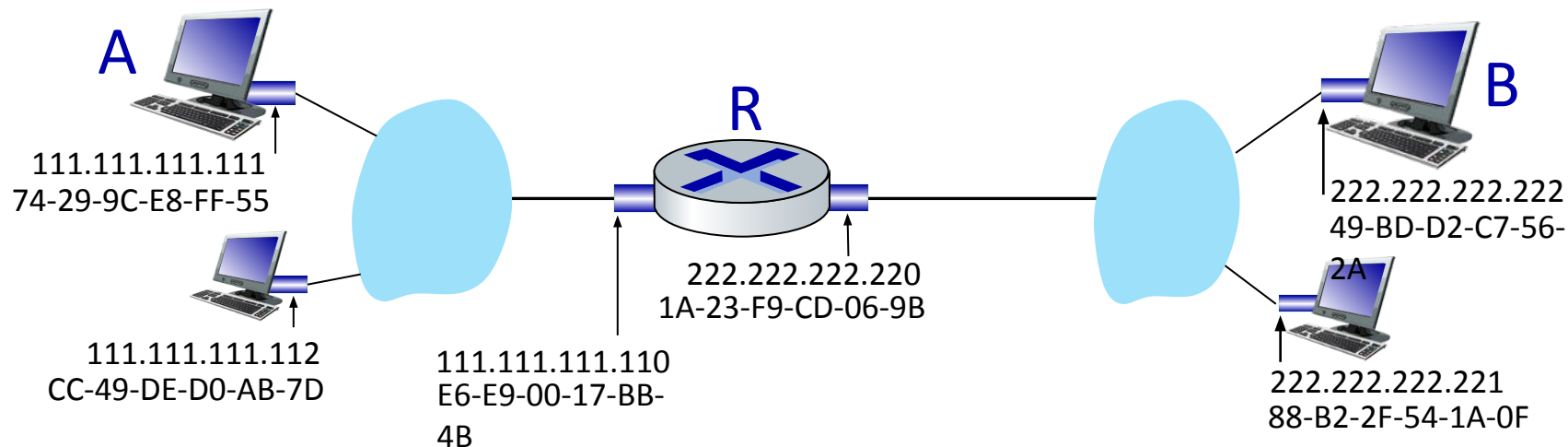


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Routing to another Subnet : Addressing

Walkthrough: sending a datagram from *A* to *B* via *R*

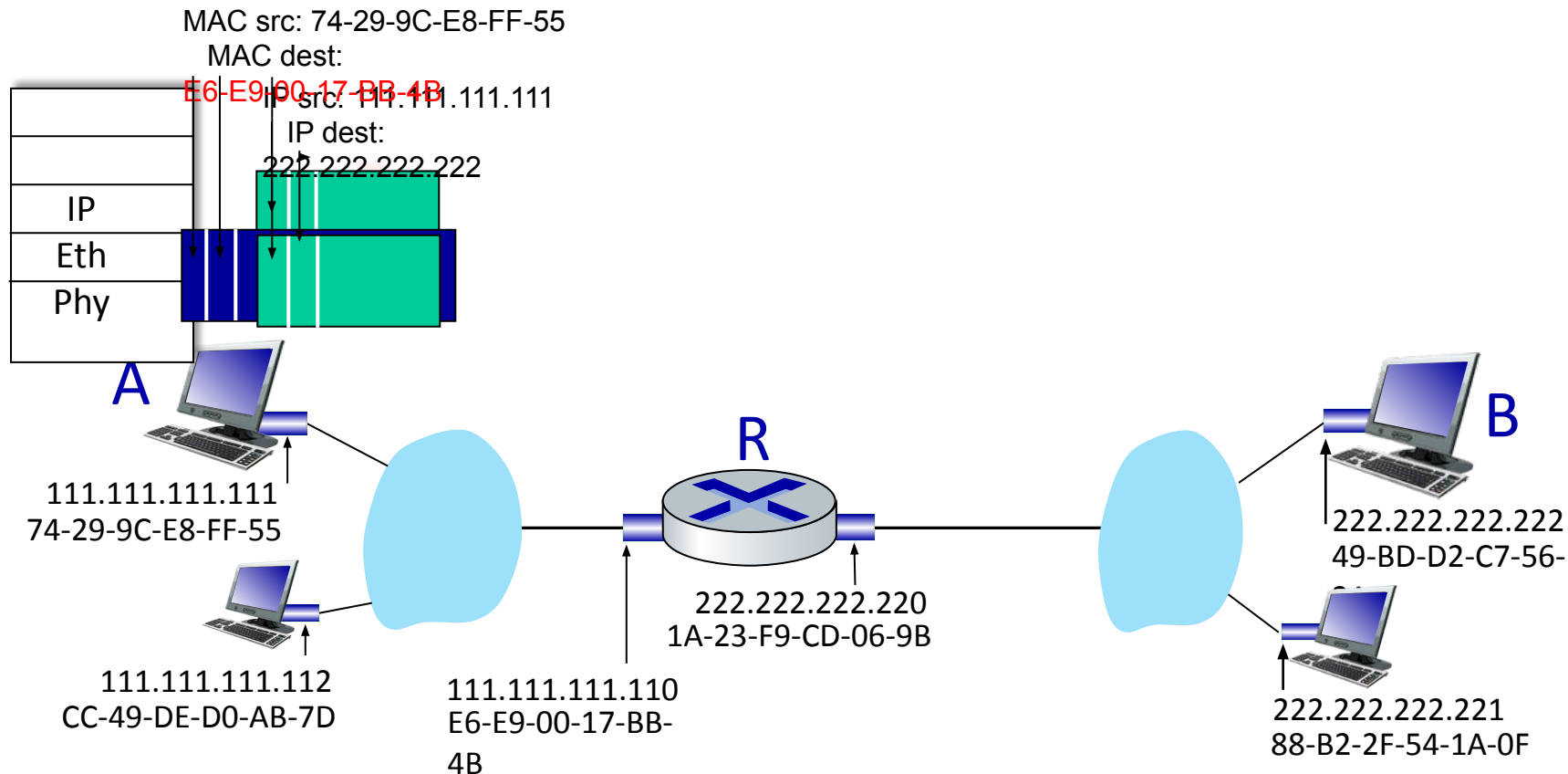
- Focus on addressing – at IP (datagram) and MAC layer (frame) levels
- Assume that:
 - A knows B's IP address
 - A knows IP address of first hop router, R (how?)
 - A knows R's MAC address (how?)



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Routing to another Subnet : Addressing

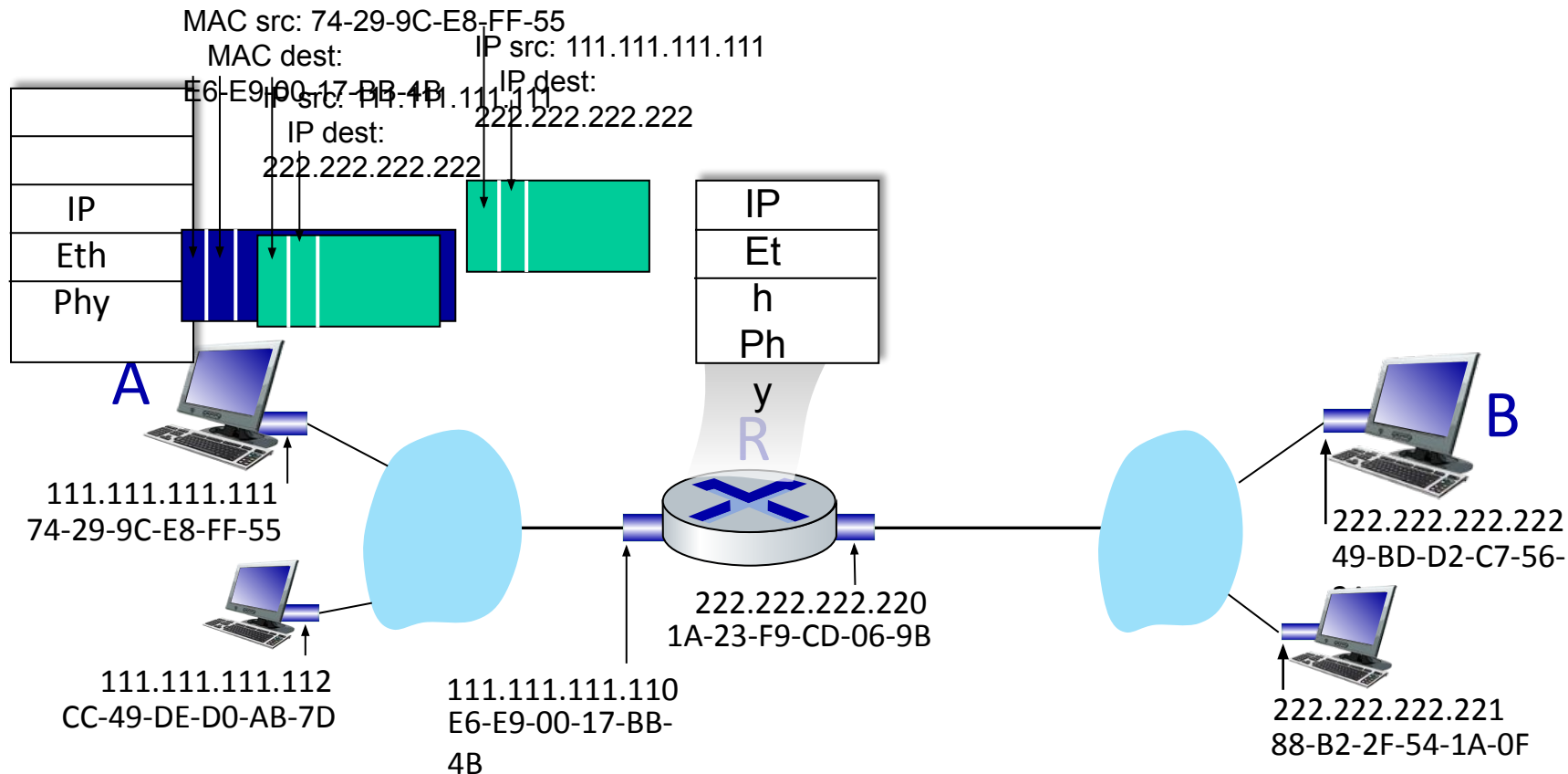
- A creates IP datagram with IP source A, destination B
- A creates link-layer frame containing A-to-B IP datagram
 - R's MAC address is frame's destination



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Routing to another Subnet : Addressing

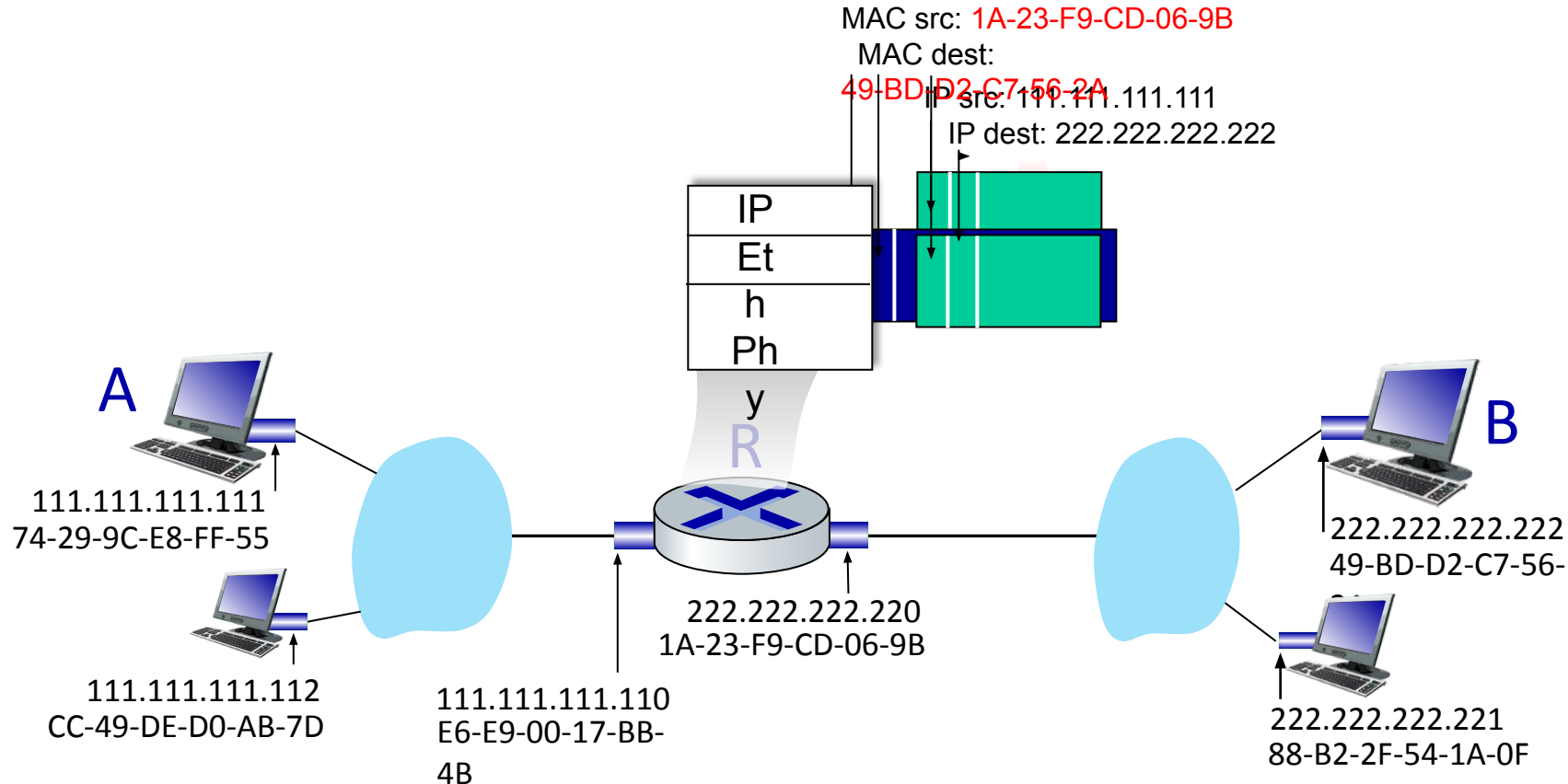
- frame sent from A to R
- frame received at R, datagram removed, passed up to IP



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Routing to another Subnet : Addressing

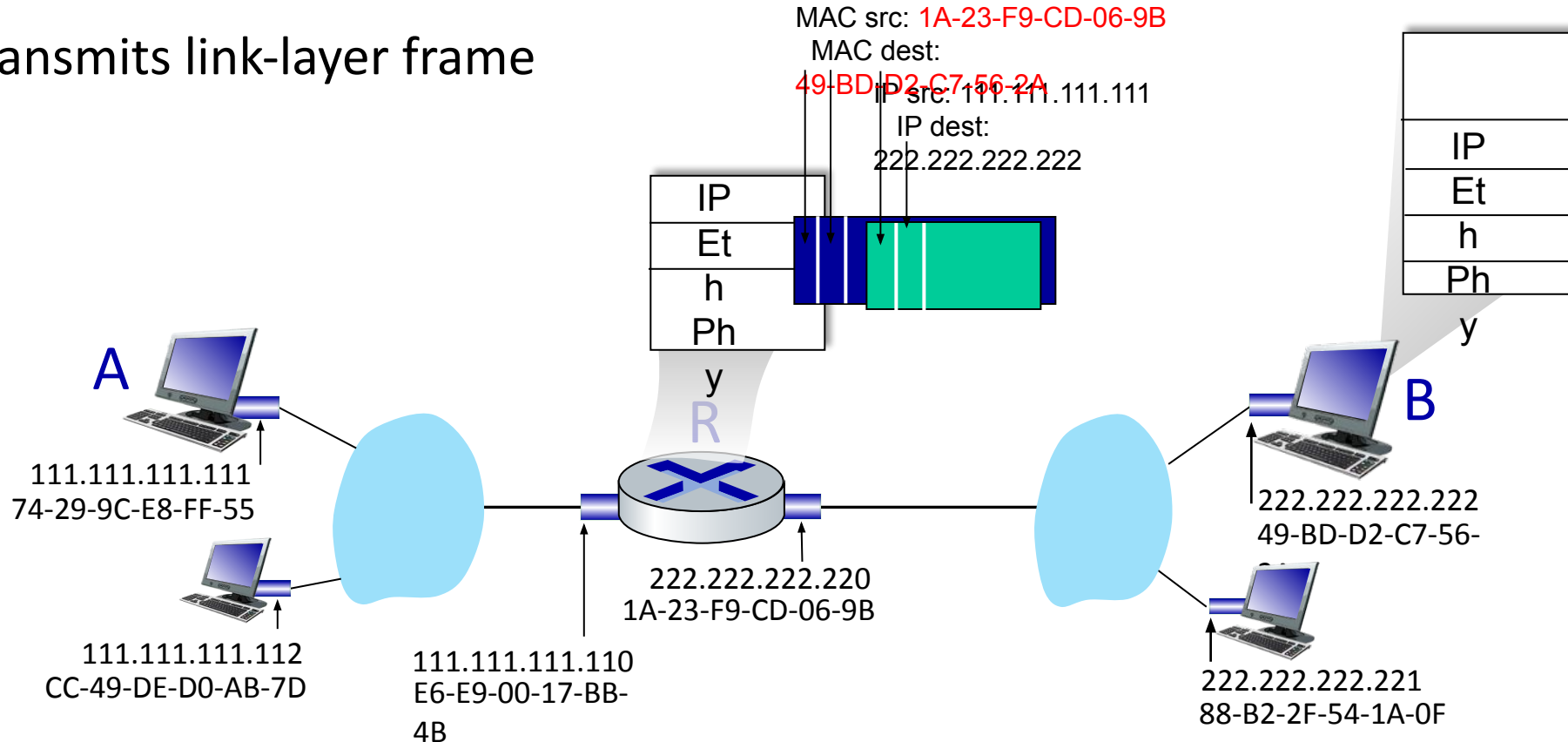
- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address



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Routing to another Subnet : Addressing

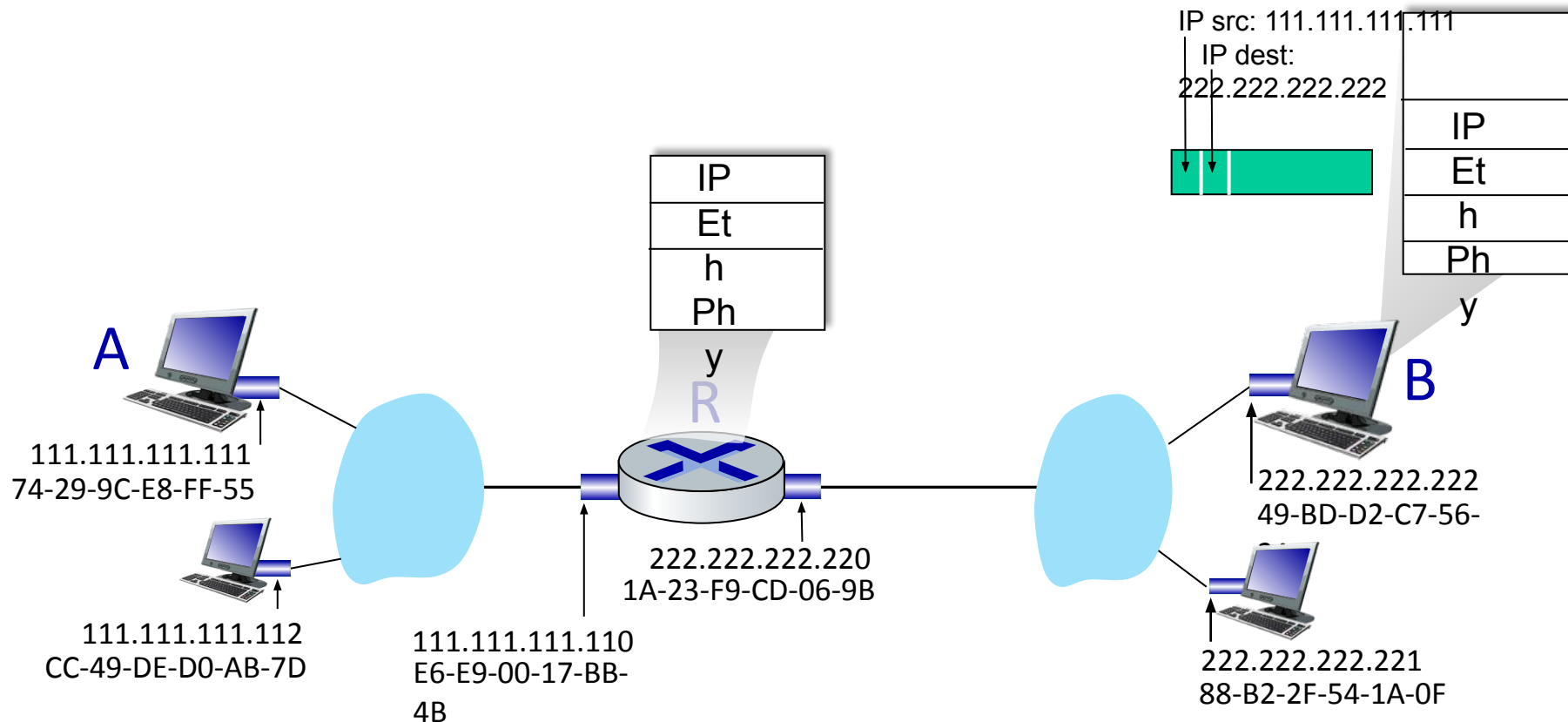
- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address
- Transmits link-layer frame



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Routing to another Subnet : Addressing

- B receives frame, extracts IP datagram destination B
- B passes datagram up protocol stack to IP



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Unit – 4 Link Layer and LAN Roadmap



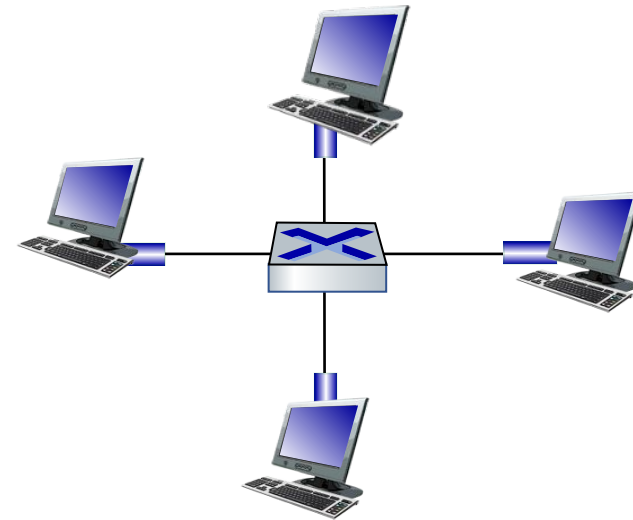
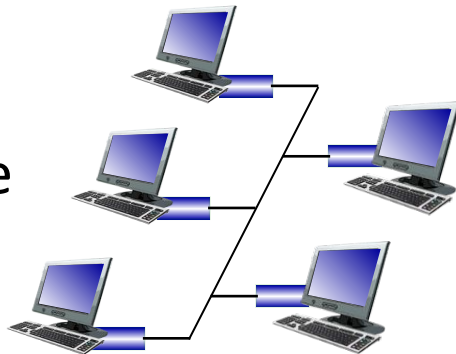
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“dominant” wired LAN technology:

- first widely used LAN technology
- simpler, cheap
- kept up with speed race: 10 Mbps – 400 Gbps
- single chip, multiple speeds (e.g., Broadcom BCM5761)
- Ethernet’ Success
 - First widely deployed high-speed LAN
 - Token ring, FDDI, and ATM were more complex and expensive than Ethernet
 - Operated at equal data rates or higher.

- **Bus:** popular through mid 90s
 - all nodes in same collision domain (can collide with each other)
- **Switched:** prevails today
 - active link-layer 2 *switch* in center
 - each “spoke” runs a (separate) Ethernet protocol (nodes do not collide with each other)

bus: coaxial cable



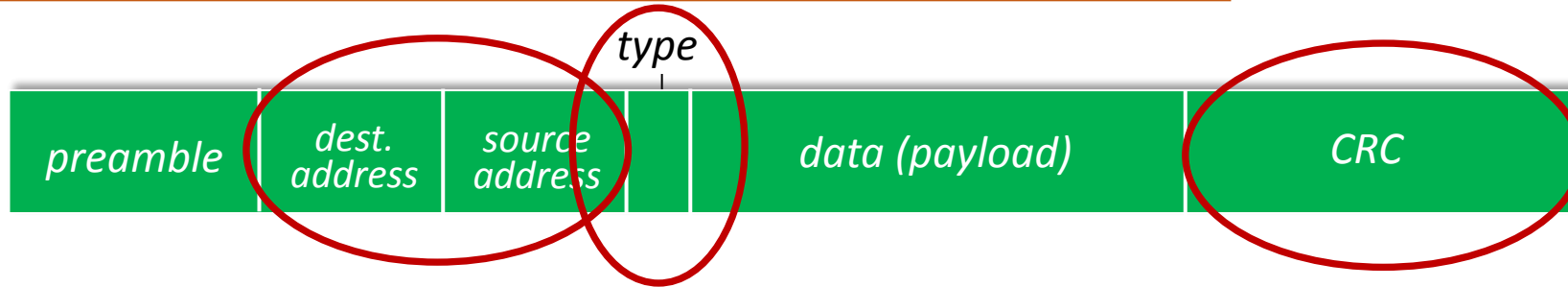
switched

Sending interface encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



Preamble:

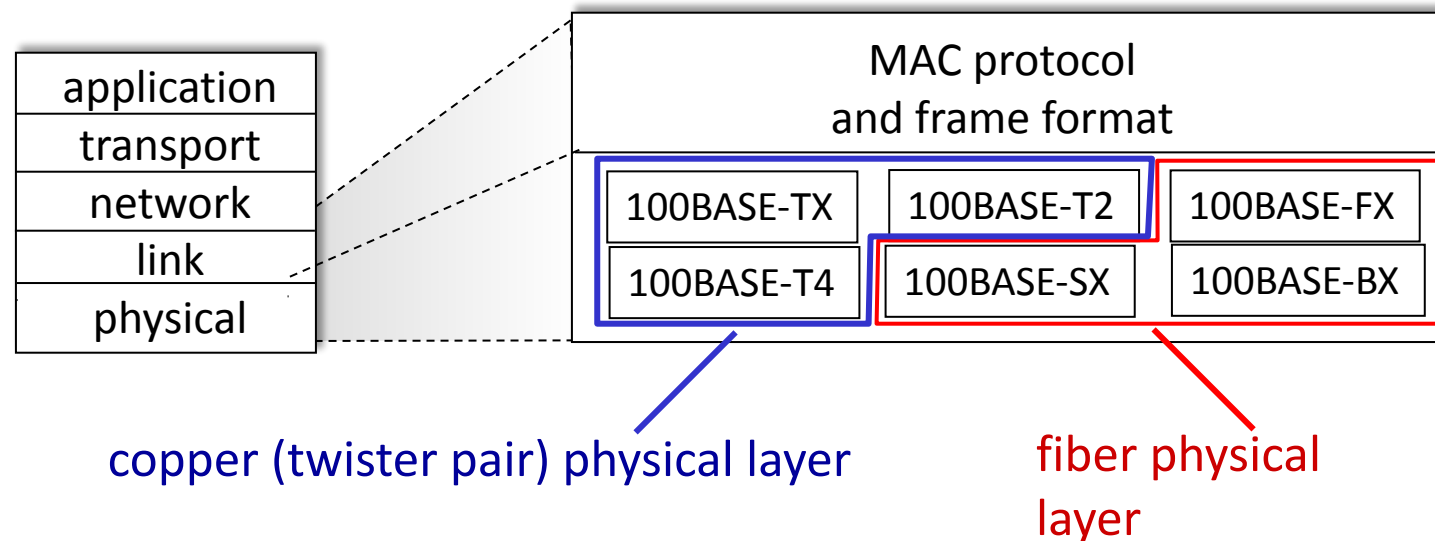
- Used to synchronize receiver, sender clock rates
- 7 bytes of 10101010 followed by one byte of 10101011



- **Addresses:** 6 byte source, destination MAC addresses
 - if adapter receives frame with matching destination address, or with broadcast address (e.g., ARP packet), it passes data in frame to network layer protocol
 - otherwise, adapter discards frame
- **Type:** indicates higher layer protocol
 - mostly IP but others possible, e.g., Novell IPX, AppleTalk
 - used to demultiplex up at receiver
- **CRC:** cyclic redundancy check at receiver
 - error detected: frame is dropped

- **Connectionless:** no handshaking between sending and receiving NICs
- **Unreliable:** receiving NIC doesn't send ACKs or NAKs to sending NIC
 - data in dropped frames recovered only if initial sender uses higher layer rdt (e.g., TCP), otherwise dropped data lost
- Ethernet's MAC protocol: unslotted **CSMA/CD with binary backoff**

- *Many* different Ethernet standards
 - Common MAC protocol and frame format
 - Different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1Gbps, 10 Gbps, 40 Gbps
 - Different physical layer media: fiber, cable



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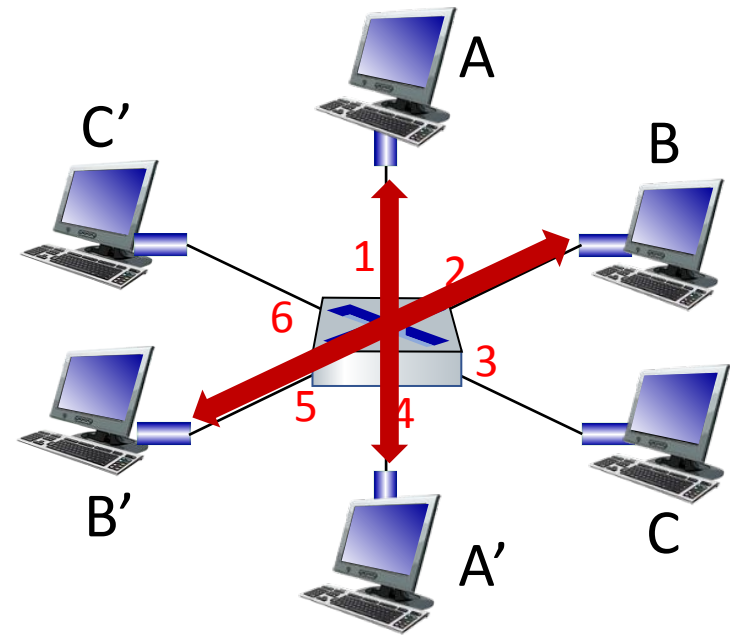
Ethernet switch

- Switch is a **link-layer** device: takes an *active* role
 - Store, forward Ethernet frames
 - Examine incoming frame's MAC address, *selectively* forward frame to one-or-more outgoing links when frame is to be forwarded on segment, uses CSMA/CD to access segment
- **Transparent:** hosts *unaware* of presence of switches
- **Plug-and-play, self-learning**
 - Switches do not need to be configured

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Switch : Multiple Simultaneous Transmissions

- Hosts have dedicated, direct connection to switch
- Switches buffer packets
- Ethernet protocol used on *each* incoming link, so:
 - no collisions; full duplex
 - each link is its own collision domain
- **Switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions

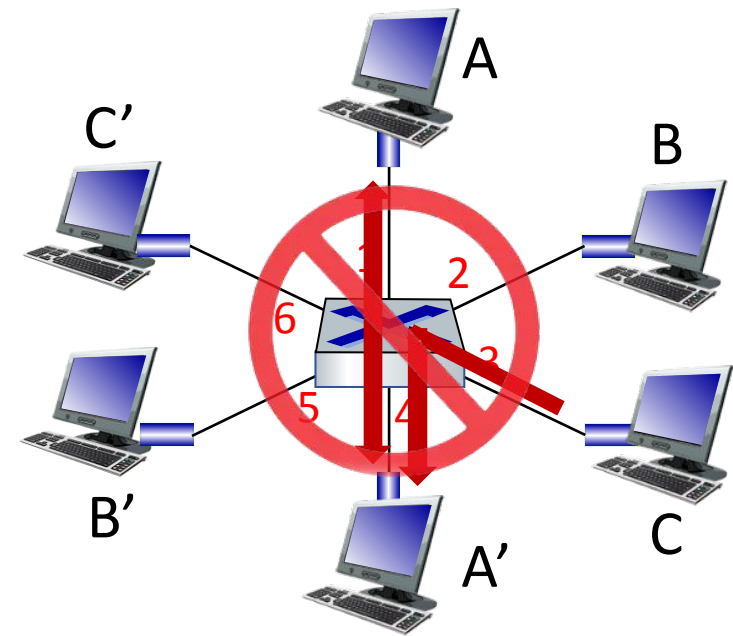


switch with six
interfaces (1,2,3,4,5,6)

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Switch: Multiple Simultaneous Transmissions

- Hosts have dedicated, direct connection to switch
- Switches buffer packets
- Ethernet protocol used on *each* incoming link, so:
 - No collisions; full duplex
 - Each link is its own collision domain
- **Switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions
 - but A-to-A' and C to A' can *not* happen simultaneously



switch with six
interfaces (1,2,3,4,5,6)

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Switch Forwarding Table

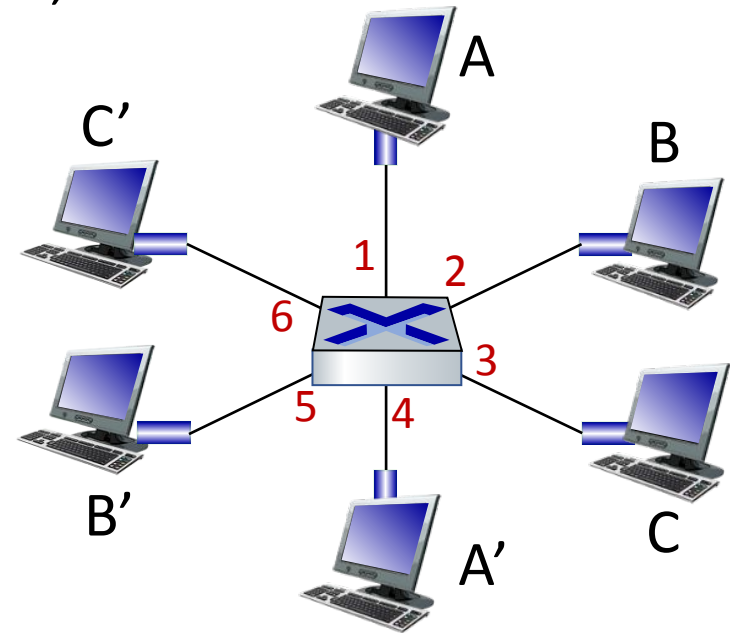
Q: How does switch know A' reachable via interface 4,
B' reachable via interface 5?

A: Each switch has a **switch table**, each entry:

- (MAC address of host, interface to reach host, time stamp)
- looks like a routing table!

Q: How are entries created, maintained in switch table?

- something like a routing protocol?



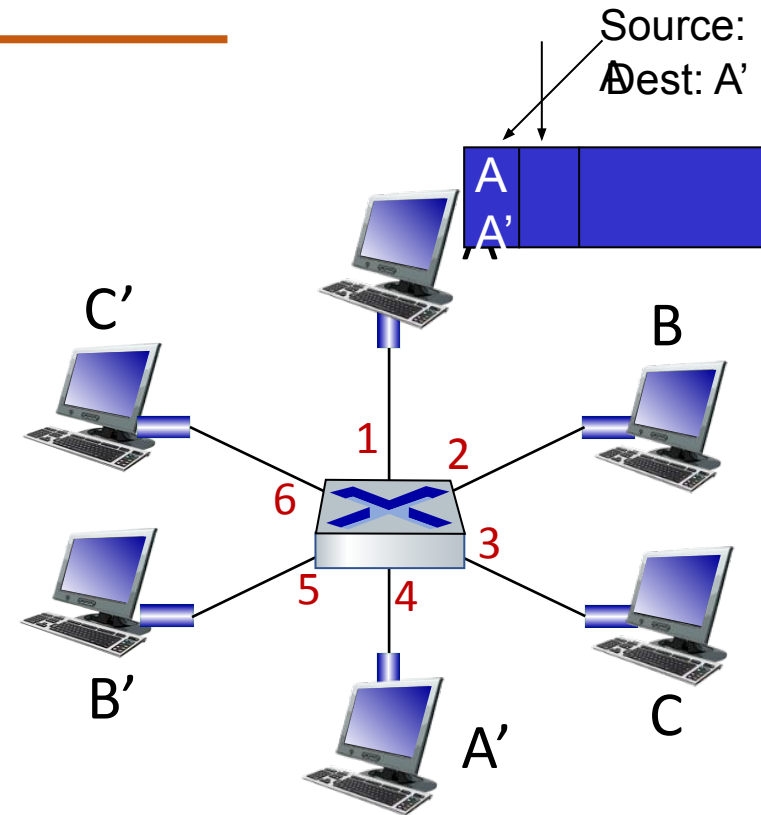
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Switch: Self-learning

- Switch *learns* which hosts can be reached through which interfaces
 - When frame received, switch “learns” location of sender: incoming LAN segment
 - Records sender/location pair in switch table

*Switch table
(initially empty)*

MAC addr	interface	
TTL		
A	1	60



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Switch: Frame Filtering / Forwarding

When frame received at switch:

1. Record incoming link, MAC address of sending host
2. Index switch table using MAC destination address
3. If entry found for destination
then {
 If destination on segment from which frame arrived
 then drop frame
 else forward frame on interface indicated by entry
}
else flood /* forward on all interfaces except arriving interface */

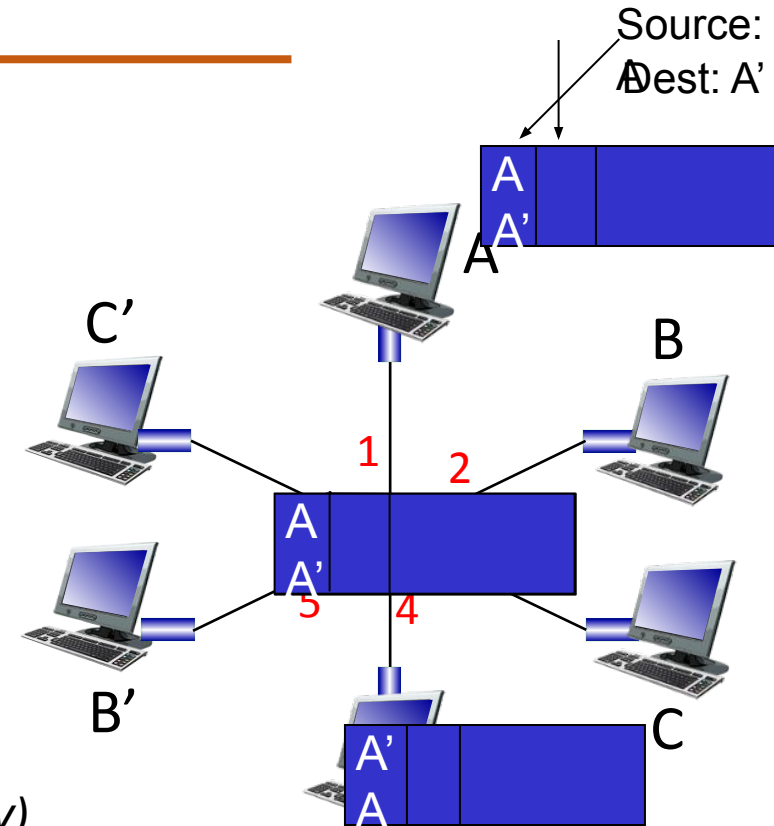
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Self-learning, Forwarding: Example

- Frame destination, A', location unknown: **Flood**
- Destination A location known: **Selectively send on just one link**

MAC addr	interface	
TTL		
A	1	60
A'	4	60

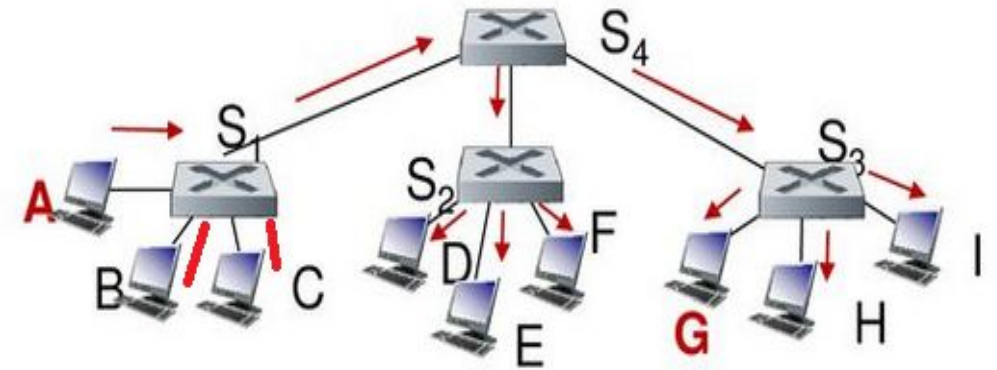
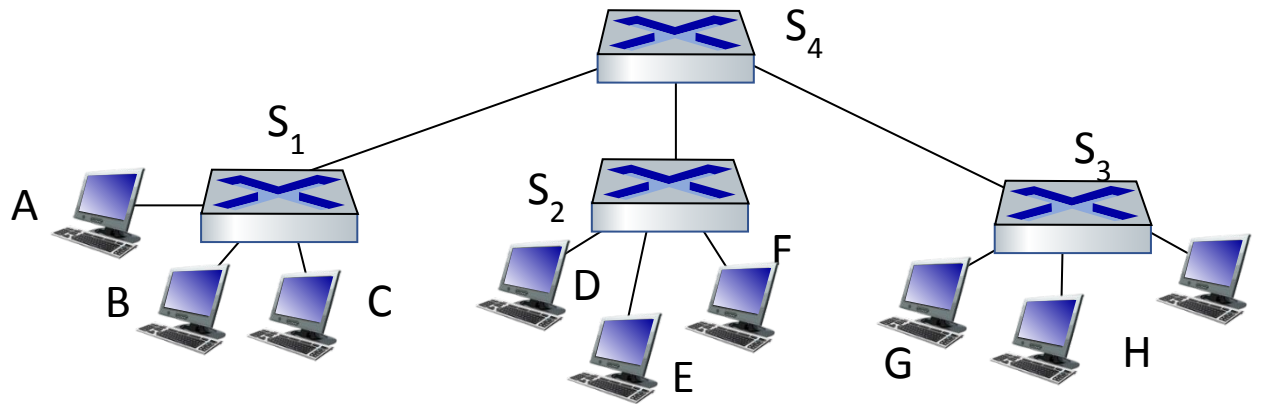
*switch table
(initially empty)*



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Interconnecting Switches

Self-learning switches can be connected together:



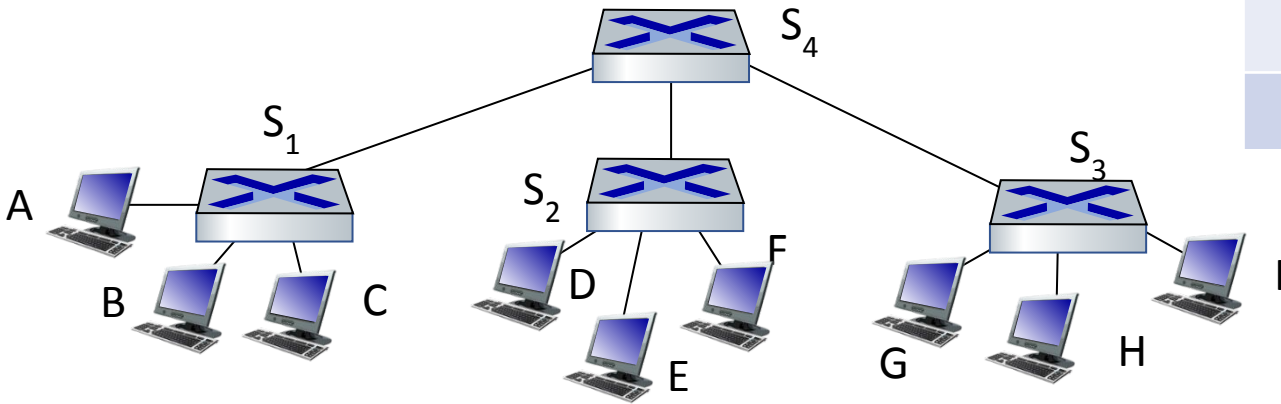
Q: sending from A to G - how does S₁ know to forward frame destined to G via S₄ and S₃?

- A: self learning! (works exactly the same as in single-switch case!)

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Self-learning Multi-switch Example

Suppose C sends frame to I, I responds to C



Switch 1	
Address	Port
C	1
I	4

Switch 4	
Address	Port
C	1
I	3

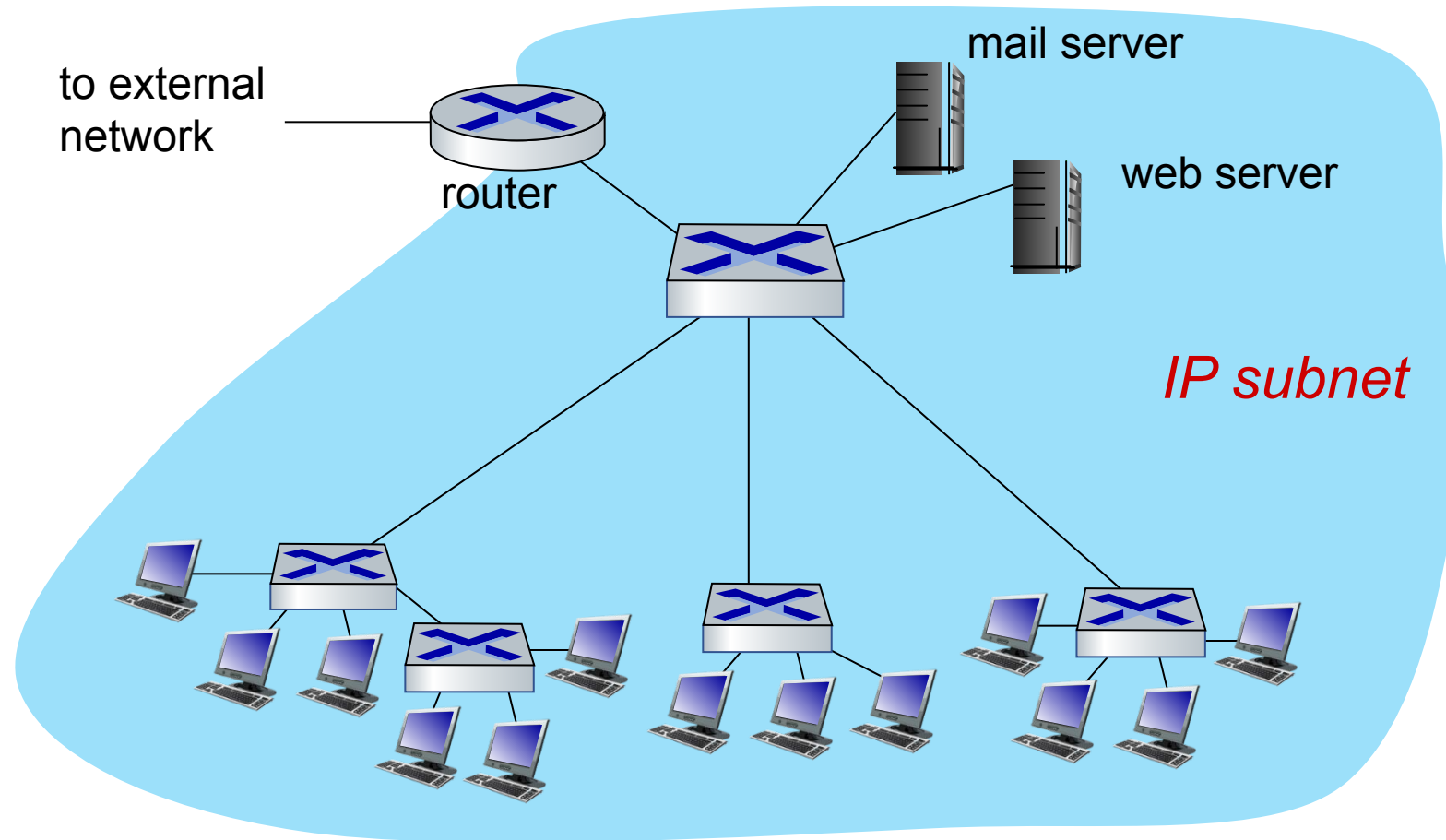
Switch 2	
Address	Port
C	4

Switch 3	
Address	Port
C	4
I	3

Q: show switch tables and packet forwarding in S₁, S₂, S₃, S₄

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Small Institutional Network



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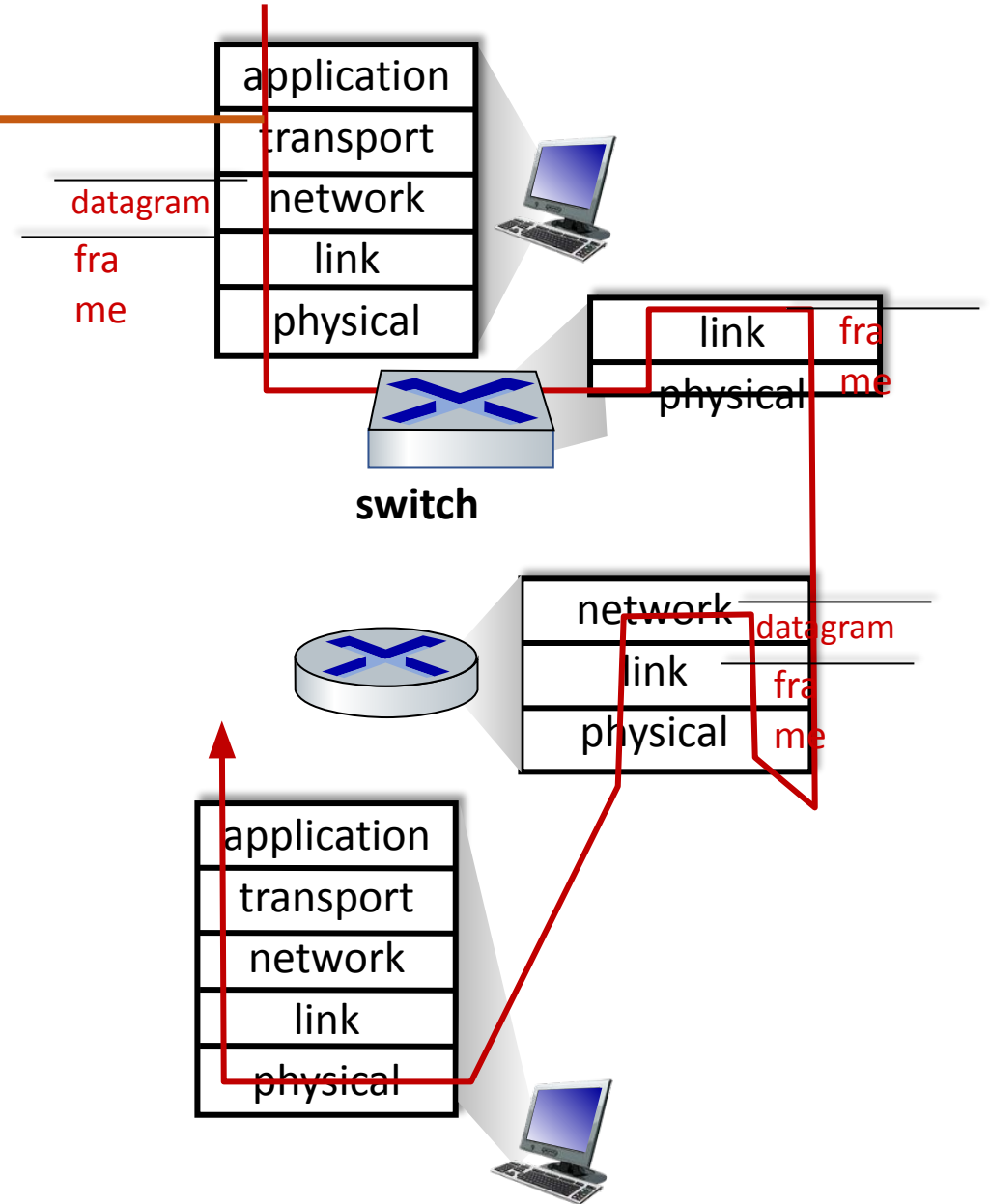
Switches Vs Routers

Both are store-and-forward:

- **Routers:** network-layer devices (examine network-layer headers)
- **Switches:** link-layer devices (examine link-layer headers)

Both have forwarding tables:

- **Routers:** compute tables using routing algorithms, IP addresses
- **Switches:** learn forwarding table using flooding, learning, MAC addresses





THANK YOU

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